Class XII Session 2025-26
Subject - Physics
Sample Question Paper - 4

Time Allowed: 30 minutes

General Instructions:

1. There are 33 questions in all. All questions are compulsory.

   This question paper has five sections: Section A,

2. Use of calculators is not allowed.

Section A

1. Hole is **[1]**

   a) A vacancy created when an electron leaves a covalent bond.
   b) Gap between valence band and conduction band
   c) Particle similar to that of electron
   d) An anti-particle of electron.

2. Thermo emf ε (in volts) of a certain thermocouple is found to vary with θ (in ^{o}C) according to equation E = $20\theta - \frac{\theta^2}{20}$, where θ is the temperature of the hot function, the cold function being kept at 0° C. Then, the neutral temperature of the thermocouple is: **[1]**

   a) 100°C
   b) 340°C
   c) 200°C
   d) 300°C

3. For a glass prism, the angle of minimum deviation will be smallest for the light of **[1]**

   a) blue colour
   b) yellow colour
   c) red colour.
   d) green colour

4. The susceptibility of a magnetic substance is found to depend on temperature and the strength of the magnetising field. The material is a: **[1]**

   a) diamagnet
   b) superconductor

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c) paramagnet

d) ferromagnet

5. A series combination of n_1 capacitors, each of value C_1 is charged by a source of potential difference 4V. When another parallel combination of n_2 capacitors, each of value C_2, is charged by a source of potential difference V, it has the same (total) energy stored in it, as the first combination has. The value of C_2, in terms of C_1, is then: **[1]**

a) $\frac{2C_1}{n_1 n_2}$

b) $16\frac{n_2}{n_1}C_1$

c) $2\frac{n_2}{n_1}C_1$

d) $\frac{16C_1}{n_1 n_2}$

6. Magnetic field due to a straight solenoid at any point inside it is B = μ_0 ni. Magnetic field at the end of the solenoid is **[1]**

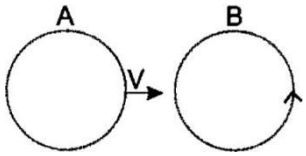
a) $\frac{B}{4}$

b) 2B

c) $\frac{B}{2}$

d) B

7. There are two coils A and B as shown in the figure. A current starts flowing in B as shown, when A is moved towards B and stops when A stops moving. The current in A is counter clockwise. B is kept stationary when A moves. We can infer that **[1]**



a) there is a constant current in the counterclockwise direction in A.

b) there is a constant current in the clockwise direction in A.

c) there is no current in A.

d) there is a varying current in A.

8. The intensity of magnetic field is H and the moment of a magnet is M Maximum potential energy is: **[1]**

a) 4 MH

b) MH

c) 2 MH

d) 3 MH

9. The maximum number of possible interference maxima for slit separation equal to twice the wavelength in Young's double-slit experiment is: **[1]**

a) three

b) zero

c) five

d) infinite

10. Two point charges q_1 and q_2 are at separation r. The force acting between them is given by $F = K\frac{q_1 q_2}{r^2}$. The constant K depends upon **[1]**

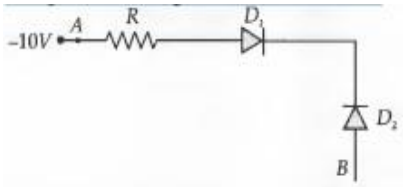
a) neither on only on the system of units nor on only on medium between charges

b) only on medium between charges

c) only on the system of units

d) both on only on the system of units and only on medium between charges

11. In the given Fig., assuming the diodes to be ideal, **[1]**



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a) D_1 and D_2 are both forward biased and hence current flows from A to B.

b) D_2 is forward biased and D_1 is reverse biased and hence no current flows from B to A and vice versa.

c) D_1 and D_2 are both reverse biased and hence no current flows from A to B and vice versa.

d) D_1 is forward biased and D_2 is reverse biased and hence current flows from A to B

12. The relationship between angle of incidence i, prism of angle A and angle of minimum deviation for a triangular prism is **[1]**

a) $2A + \delta_m = i$

b) $A + \delta_m = 2i$

c) $A + \frac{\delta_m}{2} = i$

d) $A + \delta_m = i$

13. **Assertion (A):** If frequency of incident light is doubled, the kinetic energy of photoelectron is also doubled. **[1]**
Reason (R): The kinetic energy of photoelectron is directly proportional to frequency of incident light.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

14. **Assertion:** If a conducting medium is placed between two charges, then electric force between them becomes zero. **[1]**
Reason: Reduction in a force due to introduced conductor is proportional to its dielectric constant.

a) Assertion and reason both are correct statements and reason is correct explanation for assertion.

b) Assertion and reason both are correct statements but reason is not correct explanation for assertion.

c) Assertion is correct statement but reason is wrong statement.

d) Assertion is wrong statement but reason is correct statement.

15. **Assertion (A):** When a tiny circular obstacle is placed in the path of light from a distant source, a bright spot is seen at the centre of the shadow of the obstacle. **[1]**
Reason (R): Wave diffracted from the edges of circular obstacle interfere constructively at the centre of the shadow resulting in the formation of a bright spot.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

16. **Assertion (A):** An alternating current does not show any magnetic effect. **[1]**
Reason (R): Alternating current varies with time.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

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